

Junhyung Ahn

Ph.D. in Information Sciences • University of Illinois Urbana–Champaign (UIUC)

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RESEARCH INTERESTS

Causal inference; Agentic AI; Robust machine learning.

EDUCATION

University of Illinois Urbana–Champaign (UIUC)

Aug. 2026 – Present

Ph.D. in Information Sciences

Advisor: Prof. Yonghan Jung

Korea Advanced Institute of Science and Technology (KAIST)

Feb. 2019 – Feb. 2022

M.S. in Electrical Engineering

Advisor: Prof. Changho Suh

Thesis: Leveraging Hierarchical Similarity Graphs for Matrix Completion

Korea Advanced Institute of Science and Technology (KAIST)

Mar. 2015 – Feb. 2019

B.S. in Electrical Engineering

Magna Cum Laude (GPA 3.93/4.3)

PUBLICATIONS

PREPRINT

1. **Junhyung Ahn**. “AGORA: Reliability-Aware Aggregation of LLM Judges via Item-Response Theory.” *In preparation*.

PEER-REVIEWED

1. **Junhyung Ahn** and Sanghack Lee. “On Predicting Post-Click Conversion Rate via Counterfactual Inference.” *IEEE International Conference on Data Mining (ICDM)*, 2025. **Best Paper Award Finalist**
2. **Junhyung Ahn**, Adel Elmahdy, Soheil Mohajer, and Changho Suh. “On the Fundamental Limits of Matrix Completion: Leveraging Hierarchical Similarity Graphs.” *IEEE Transactions on Information Theory*, 70(3):2039–2075, 2024.
3. Adel Elmahdy*, **Junhyung Ahn***, Soheil Mohajer, and Changho Suh. “The Optimal Sample Complexity of Matrix Completion with Hierarchical Similarity Graphs.” *IEEE International Symposium on Information Theory (ISIT)*, pp. 2409–2414, 2022.
4. Adel Elmahdy*, **Junhyung Ahn***, Changho Suh, and Soheil Mohajer. “Matrix Completion with Hierarchical Graph Side Information.” *Conference on Neural Information Processing Systems (NeurIPS)*, pp. 9061–9074, 2020.

(* equal contribution.)

RESEARCH EXPERIENCE

AI Engineer, Agent Flow Modeling Team

Aug. 2025 – Aug. 2026

NAVER • Seongnam, Republic of Korea

- Developed off-policy evaluation methods for deterministic eCPM-based ranking policies in online ad auctions, using kernel-based relaxation to reduce variance in importance-sampling estimators.
- Designed an LLM-as-a-Judge evaluation framework for automated quality assessment of AI-generated search responses, including bias mitigation strategies and an ensemble judging pipeline to replace human annotation at scale

Machine Learning Engineer, Artificial Intelligence Team

Mar. 2022 – May 2025

Dable • Seoul, Republic of Korea

- Proposed a causal-inference approach to conversion-rate (CVR) prediction that imputes conversion labels for non-clicked samples via counterfactual inference, mitigating selection bias under low conversion rates; published at IEEE ICDM.
- Addressed data scarcity with multi-task and self-supervised learning (feature-embedding augmentation, contrastive objectives) to improve CVR estimation under data-limited conditions.

Graduate Researcher, Information Systems Lab

Feb. 2019 – Feb. 2022

KAIST • Daejeon, Republic of Korea

- Characterized the information-theoretic limit on observed entries required for perfect matrix recovery, and designed an efficient algorithm using hierarchical social-similarity graphs as side information to address the cold-start problem (NeurIPS, ISIT, IEEE Trans. Inf. Theory).
- Built a deep retrieval system for chest-disease prediction (collaboration with Samsung Medical Center), classifying 14 disease types with semantically meaningful feature representations.

HONORS & AWARDS

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| • Best Paper Award Finalist, IEEE ICDM | 2025 |
| • Magna Cum Laude, KAIST | 2019 |
| • KAIST Support Scholarship | 2019–2020 |
| • KAIST Support Scholarship | 2015–2018 |
| • National Science & Technology Scholarship | 2015–2018 |

TEACHING EXPERIENCE

Instructor, Industry AI Courses

Aug. 2019 – Dec. 2021

Taught applied machine-learning courses for industry engineers (Samsung, SK Hynix, Hyundai Motor, Seongnam–KAIST program).

Teaching Assistant

Sep. 2019 – Aug. 2020

Wireless Communication Systems (EE321) and Introduction to Electronics Design Lab (EE305).